Stream separation improves Bregman conditioning in transformers

J. Clayton Kerce
Georgia Tech Research Institute
clayton.kerce@gtri.gatech.edu

Abstract

Linear methods for steering transformer representations, including probing, activation engineering, and concept erasure, implicitly assume the geometry of representation space is Euclidean. Park et al. [Park et al., 2026] showed that softmax induces a curved Bregman geometry whose metric tensor is the Hessian of the log-normalizer, $H(\lambda) = \mathrm{Cov}[\gamma \mid \lambda]$. Ignoring this curvature causes Euclidean steering to leak probability mass to unintended tokens. Their analysis applies at the output layer. We measure this Hessian at intermediate layers in a controlled 2×2 design crossing stream separation with per-layer supervision (vocabulary decoding loss at each layer), all at matched vocabulary and parameter count. In standard single-stream transformers, H is severely degenerate at intermediate layers (effective rank 8 in 516 dimensions). Stream separation improves conditioning by up to $22\times$ in effective rank, even without auxiliary supervision. Per-layer supervision helps, but less. The cosine similarity between primal and dual concept directions predicts per-layer steering effectiveness on downstream tasks, with a threshold near 0.3. These results bear on the reliability of linear safety interventions, which depend on the geometry being well-conditioned at the layer where they are applied.

1 Introduction

Probing, activation engineering, and concept erasure are linear methods. They identify a direction in representation space and add, remove, or read off a signal along that direction [Alain and Bengio, 2017, Turner et al., 2023, Belrose et al., 2023]. Each method assumes that moving along a straight line in representation space produces a predictable change in the output distribution. This holds when the geometry is flat. It can fail when the geometry is curved.

The output layer of a transformer produces a probability distribution over the vocabulary using the softmax function. Softmax converts logits λ into a probability distribution $p(y \mid \lambda) = \exp(\lambda^\top \gamma_y - A(\lambda))$, where $A(\lambda) = \log \sum_y \exp(\lambda^\top \gamma_y)$ is the log-normalizer, the function that controls how softmax distributes probability mass across the vocabulary. The Hessian of A is

$$H(\lambda) \;=\; \nabla^2 A(\lambda) \;=\; \mathrm{Cov}[\gamma \mid \lambda]. \tag{1}$$

Park et al. [Park et al., 2026] showed that H is the metric tensor of a dually flat Bregman geometry induced by softmax. "Dually flat" means the space has two natural coordinate systems, primal and dual, related through H. When H is well-conditioned, primal directions approximate dual directions, and linear steering works. When H is poorly conditioned, primal and dual directions diverge, and Euclidean steering leaks probability mass to unintended tokens. Their analysis applies at the output layer.

[0]This work was partially supported by DARPA Contract HR001125C0302.

Preprint.

We measure the conditioning of H at every intermediate layer. We compare two interventions in a 2×2 factorial design: stream separation (the CASCADE architecture [Kerce and Fox, 2026], which maintains a frozen token-equivalent stream alongside a general embedding stream to preserve symbolic representations throughout the network) versus single-stream, crossed with per-layer supervision (vocabulary decoding loss at the output of each feed-forward block) versus no auxiliary loss. All four models share the same vocabulary, parameter count, and training data.

We find three things.

1. In single-stream transformers without auxiliary loss, H is severely degenerate at intermediate layers: effective rank 8 in a 516-dimensional space. Linear methods at these layers operate in 2% of the geometry.
2. Stream separation improves conditioning more than the auxiliary loss. CASCADE models without auxiliary loss have better-conditioned geometry than single-stream models with it. Architecture, not training objective, is the primary determinant.
3. The cosine similarity between primal and dual concept directions predicts per-layer steering effectiveness on downstream tasks. Below 0.3, Euclidean steering is unreliable. Above 0.4, it works. This gives practitioners a pre-deployment check.

These results matter because linear safety interventions are increasingly applied at intermediate layers. A steering vector that appears to work (the probe reads the intended signal) can silently redistribute probability mass to unintended tokens if the geometry at that layer is degenerate. We provide a cheap diagnostic to detect this failure mode and show that architecture choice determines whether it arises.

2 Background

Softmax induces a Bregman geometry on the space of logits. The metric tensor of this geometry is the Hessian $H(\lambda) = \mathrm{Cov}[\gamma \mid \lambda]$ defined in (1). Park et al. [Park et al., 2026] proved that steering along the dual geodesic (the natural gradient direction under H) is optimal at the output layer: it moves probability mass to the target concept while minimizing off-target disturbance (their Theorem 3). We extend their measurement to intermediate layers, where their optimality guarantee does not directly apply.

The primal coordinates λ are the logits. The dual coordinates $\eta = \nabla A(\lambda) = \mathbb{E}[\gamma \mid \lambda]$ are the expected sufficient statistics, which for softmax are the token probabilities $p(y \mid \lambda)$. A small step in primal space maps to a step in dual space through $d\eta = H(\lambda)\, d\lambda$. A primal direction is a straight line in logit space: the direction that standard linear methods use. A dual direction is the corresponding geodesic under the Bregman metric: the direction that accounts for how softmax curves the space and moves probability mass most efficiently. When H is well-conditioned, the two approximately agree, and linear steering works. When H is ill-conditioned, some primal directions are crushed to near-zero in the dual space while others are amplified. Euclidean steering then leaks probability mass to unintended tokens.

We quantify conditioning with two measures. The *effective rank* $\mathrm{rank}_{\mathrm{eff}}(H) = \exp\!\big(-\sum_i \hat{\sigma}_i \log \hat{\sigma}_i\big)$, where $\hat{\sigma}_i = \sigma_i / \sum_j \sigma_j$ are the normalized singular values, counts how many directions in H carry meaningful variance [Roy and Vetterli, 2007]. If the effective rank is 8 in a 516-dimensional space, then 98% of the geometry is compressed into 8 directions, and a linear intervention that does not land in those directions will have unpredictable effects on the output distribution. The *condition number* $\kappa(H) = \sigma_{\max}/\sigma_{\min}$ measures the worst-case ratio of amplification to compression in the primal-to-dual mapping: how much H distorts one direction relative to another.

3 Experimental design

Models. We compare four transformer language models in a 2×2 factorial design. The first factor is architecture: *stream separation* versus *single-stream*. In the stream-separated (CASCADE) architecture [Kerce and Fox, 2026], the token embedding stream x_t is frozen after initialization and a separate contextual stream x_e carries all learned computation; logits at any layer are computed

Table 1: Effective rank of $H^{(\ell)}$ by layer. All models have 516 dimensions. Higher is better.

	CASCADE		Single-stream	
Layer	+ aux. loss	control	+ aux. loss	control
0	184	8	123	9
1	75	156	49	14
2	26	9	142	9
3	36	6	183	8
4	88	8	72	8
5	37	4	17	7
final	34	4	17	7

Table 2: Condition number $\kappa(H^{(\ell)})$ by layer. Lower is better.

	CASCADE		Single-stream	
Layer	+ aux. loss	control	+ aux. loss	control
0	1,297	14,701	2,185	49,392
1	4,956	1,200	4,956	11,238
3	5,672	19,768	1,217	20,142
5	8,481	34,275	12,095	21,131
final	8,399	34,275	8,166	21,131

from $x_t + x_e^{(\ell)}$. Because the token stream is always in the coordinate system of the embedding matrix, logits at every layer are computed in the same space as the output softmax. In the single-stream architecture, a single residual stream carries both token and contextual information, as in a standard transformer. The second factor is training objective: an auxiliary cross-entropy loss at every intermediate layer (with weight decaying linearly from 0.1 at the first layer to 0.8 at the final layer) versus standard next-token loss only. Architecture details are in appendix B.

Controls. All four models use the GPT-2 tokenizer (50,257 tokens), have 45.4M parameters, 6 layers, 6 heads, 516-dimensional embeddings, and gated attention. They are trained on the same data mixture for the same number of steps.

Hessian measurement (Phase 1). For each model and each layer ℓ, we extract pre-softmax logits at the last token position across 480 contexts (30 batches of 16). We compute the sample covariance $\hat{H}^{(\ell)} = \mathrm{Cov}[\gamma \mid \lambda]$ using a top-20K token approximation and report the effective rank, condition number, and trace at each layer.

Steering comparison (Phase 2). We construct a binary concept direction from 15 gendered word pairs (e.g., *king/queen*, *boy/girl*) using 40 gendered prompts. From 10 starting contexts per layer, we steer representations toward the target concept using both Euclidean addition and dual (natural gradient) steering following the protocol of Park et al. [Park et al., 2026]. We measure off-target KL divergence at the step where target concept probability first reaches 80%. The difference in off-target KL between the two methods quantifies how much the geometry penalizes Euclidean steering at each layer. In practical terms, this measures collateral damage: how much the rest of the output distribution is disturbed while achieving the steering goal.

4 The Hessian at intermediate layers

The central finding is stark: in single-stream transformers, the Hessian is severely degenerate at intermediate layers, while stream separation largely fixes this. Table 1 reports the effective rank at each layer for all four models, table 2 reports the condition number, and fig. 1 shows the effective rank visually.

The baseline is degenerate. The single-stream control column of table 1 shows effective rank between 7 and 14 at every layer. The Hessian has 516 dimensions, so only about 2% of directions carry

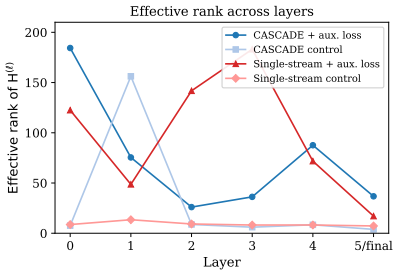

Figure 1: Effective rank of $H^{(\ell)}$ across layers. Single-stream control (pink diamonds) remains near 8. Stream separation and auxiliary loss raise it by up to $22\times$.

meaningful variance. Any linear method operating at these layers is working in a small subspace of the Bregman geometry.

Stream separation fixes this. The CASCADE control column shows effective rank between 4 and 156, with substantially higher values at layers 0 and 1. More importantly, the condition number (table 2) drops by 2–7$\times$ relative to single-stream control at layers 0 and 3–5. This improvement requires no auxiliary loss. The only difference is the frozen token stream.

The key comparison is CASCADE control versus single-stream with auxiliary loss. At deep layers (3–5 and final), CASCADE control has condition numbers of $19{,}768$–$34{,}275$, while single-stream with auxiliary loss reaches $1{,}217$–$12{,}095$. The auxiliary loss produces better condition numbers at these layers. However, the effective rank tells a different story: CASCADE control maintains higher effective rank than single-stream with auxiliary loss at layers 4–5 and final. The two measures capture different aspects of conditioning, and neither intervention dominates across all layers.

Auxiliary loss helps, but less. Adding auxiliary loss to the single-stream architecture raises effective rank from 7–14 to 17–183, an improvement of 2–22$\times$. This confirms that auxiliary supervision encourages better-conditioned geometry. But it does not close the gap with stream separation entirely. At deep layers where distributions concentrate, the single-stream auxiliary loss model still shows effective rank 17, while CASCADE with auxiliary loss reaches 34–88.

Distribution concentration. The trace of H (total variance of the softmax distribution) explains why deep layers behave differently from early layers. When the trace is near zero, the model is already confident about its prediction and there is little probability mass left to steer. In single-stream models, the trace collapses from 0.5 at layer 0 to 0.03–0.06 at layers 3–5: the softmax distribution concentrates on a few tokens early in the network. CASCADE models maintain traces of 0.4–0.7 throughout. This concentration affects the practical relevance of conditioning: at layers where distributions are already peaked, both good and bad geometry lead to similar steering outcomes, because there is little probability mass left to misallocate. Figure 2 shows this pattern.

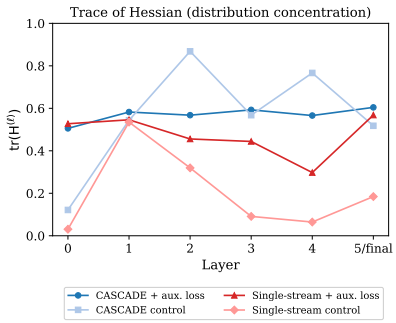

Figure 2: Trace of $H^{(\ell)}$ across layers. Single-stream control collapses at layers 3–4. CASCADE maintains higher trace, preserving probability spread.

Table 3: KL advantage of dual over Euclidean steering ($\mathrm{KL}_{\text{Euclid}} - \mathrm{KL}_{\text{dual}}$ at 80% target probability). Positive values mean Euclidean steering causes unnecessary side effects that dual steering avoids.

	CASCADE		Single-stream	
Layer	+ aux. loss	control	+ aux. loss	control
0	$+1.29$	-0.59	$+0.07$	$+5.05$
1	$+1.51$	-0.05	$+0.53$	$+3.19$
2	-2.17	$+0.04$	-0.24	$+4.33$
3	-0.56	$+0.15$	$+0.16$	$+0.59$
4	-1.75	$+0.11$	-1.00	-0.07
5	-0.76	-0.31	-0.47	-0.15
final	-0.47	-0.31	-0.44	-0.15

Table 4: Cosine similarity between primal and dual concept directions. Values below 0.3 correspond to layers where dual steering outperforms Euclidean in table 3.

	CASCADE		Single-stream	
Layer	+ aux. loss	control	+ aux. loss	control
0	0.58	0.42	0.18	0.09
2	0.55	0.48	0.17	0.09
4	0.62	0.58	0.41	0.30
5	0.64	0.60	0.57	0.52
final	0.64	0.60	0.57	0.52

5 Consequences for steering

The conditioning differences in section 4 predict that Euclidean steering should fail where the Hessian is degenerate. We test this prediction.

Where conditioning is bad, Euclidean steering fails. Table 3 shows the KL advantage of dual over Euclidean steering at each layer. In the single-stream control model, dual steering outperforms Euclidean at layers 0–3 by $+0.6$ to $+5.1$ KL. These are the layers where the Hessian is degenerate (table 1: effective rank 8–14). Euclidean steering at these layers leaks probability mass to unintended tokens.

In CASCADE models, the advantage largely vanishes. The geometry is well-conditioned enough that the simpler Euclidean method works. In CASCADE with auxiliary loss, the KL advantage is occasionally negative (Euclidean slightly outperforms dual), consistent with near-flat geometry where the overhead of computing the dual direction adds noise without benefit.

At deep layers (4–final) in all models, both methods perform similarly. The absolute KL values are small (< 0.2 for control, < 1.0 for auxiliary loss models). As noted in section 4, distributions are concentrated at these layers, leaving little probability mass to misallocate.

A cosine diagnostic predicts steering validity. The cosine similarity between the primal concept direction and the corresponding dual direction tracks the regime boundary. The intuition is direct: when primal and dual point in roughly the same direction, the curvature is mild and Euclidean steering is a good approximation. When they diverge, the curvature is severe and Euclidean steering sends probability mass in the wrong direction. Table 4 reports this quantity.

Where $\cos(\text{primal}, \text{dual}) < 0.3$, Euclidean steering is unreliable: the primal direction does not approximate the dual geodesic, and steering leaks probability mass. Where $\cos(\text{primal}, \text{dual}) > 0.4$, the two directions are aligned enough that Euclidean steering works. This pattern holds across all four models and all layers.

The diagnostic is cheap to compute (one eigendecomposition of H) and requires no steering experiment. It can be evaluated before choosing a steering method or layer. Figure 3 shows the relationship between the cosine diagnostic and the KL advantage across all models and layers.

Downstream validation. We validate the cosine diagnostic against direct vocabulary steering on four tasks (coreference resolution, induction, recency, and capitalization) using the same single-stream and CASCADE models with layer×scale sweeps across six step sizes. Steering effectiveness concentrates at layers 4–5, where $\cos > 0.3$ in the single-stream model. At layers 0–2, where $\cos < 0.1$ in the single-stream model, effects are weaker and noisier, with occasional sign inconsistencies in the coreference task. CASCADE models show clean, sign-correct effects at all layers, consistent with $\cos > 0.4$ throughout. Scale sweeps confirm linear, sign-correct scaling across all step sizes, indicating that the dual steering advantage in table 3 is not an artifact of step size choice.

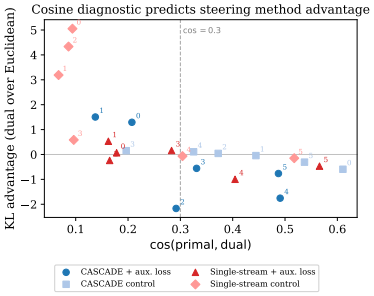

Figure 3: Cosine diagnostic vs. KL advantage. Each point is one (model, layer) pair. Left of $\cos = 0.3$ (dashed), dual wins. Right of it, both methods perform similarly.

6 Related work

Our measurement extends the information-geometric analysis of Park et al. [Park et al., 2026], who characterized the Bregman geometry of the output-layer softmax and proved that dual steering is optimal there. We ask whether their geometric structure is well-conditioned at intermediate layers, and what architectural choices affect this.

The methods whose geometric assumptions we test include linear probing [Alain and Bengio, 2017], activation addition [Turner et al., 2023], representation engineering [Zou et al., 2023], and concept erasure [Belrose et al., 2023]. Each operates by identifying or modifying a direction in representation space. Each implicitly assumes that straight lines in representation space correspond to meaningful movements in output space. Our results show that this assumption holds at some layers and architectures but not others.

Information geometry in machine learning originates with Amari's [Amari, 2016] natural gradient and the Fisher information metric. Frecon et al. [Frécon et al., 2022] studied Bregman divergences as activation functions in neural networks. Our work uses the Bregman structure that softmax induces on its inputs, following Park et al., rather than incorporating Bregman divergences into the network architecture.

Several recent results connect architecture to interpretability. Zhou et al. [Zhou et al., 2025] identified a small number of attention heads responsible for safety behavior, showing that interpretability can be localized to architectural components. Liu et al. [Liu et al., 2025] derived scaling laws for superposition, linking representational capacity to the number of features a network can encode. The CASCADE architecture [Kerce and Fox, 2026] separates token and contextual streams to improve modularity. Our contribution is to show that this separation also improves Bregman conditioning, providing a geometric account of why stream separation aids interpretability.

7 Discussion

We have established five facts. First, the softmax Hessian at intermediate layers of standard single-stream transformers is severely degenerate, with effective rank roughly 2% of the embedding dimension (table 1). Second, stream separation improves this conditioning without any change to the training objective (tables 1 and 2). Third, an auxiliary per-layer loss also improves conditioning, but less than stream separation (section 4). Fourth, where the Hessian is degenerate, Euclidean steering leaks probability mass to unintended tokens, and dual steering reduces this leakage (table 3). Fifth, the cosine similarity between primal and dual concept directions predicts which layers are geometrically safe for linear intervention, with a threshold near 0.3 (table 4). Together, these facts form a chain: degenerate geometry causes steering failure, architecture controls the geometry, and the cosine diagnostic detects the problem before deployment.

Implications for safety and assurance. Linear interventions are increasingly used to enforce behavioral constraints in deployed language models. Activation engineering adds or subtracts directions to promote refusal, reduce toxicity, or suppress unwanted capabilities. Concept erasure re-

moves directions to prevent models from representing sensitive attributes. These methods assume that the geometry of representation space is well-behaved: that a direction identified by probing corresponds to a meaningful movement in the output distribution.

Our results show that this assumption is layer-dependent and architecture-dependent. In a single-stream transformer, linear interventions at early layers operate in a 2% subspace of the Bregman geometry. A safety-critical steering vector applied at such a layer may not produce the intended behavioral change, or may produce it while simultaneously disturbing the output distribution in uncontrolled ways. The intervention appears to work in the Euclidean sense (the probe reads off the intended signal) while failing in the geometric sense (probability mass leaks to unintended tokens).

This is a verifiable failure mode, not a theoretical concern. The cosine diagnostic identifies the layers where linear interventions are geometrically valid. If $\cos(\text{primal}, \text{dual}) < 0.3$ at a given layer, a linear intervention there cannot be trusted to produce only the intended effect. This check is cheap (one eigendecomposition) and can be performed before deployment.

Stream separation provides an architectural path to reliable linear intervention. CASCADE models maintain $\cos > 0.4$ at all layers, making Euclidean steering geometrically sound throughout the network. This means that safety interventions applied at any layer will behave as intended, without uncontrolled probability leakage. The choice of architecture is therefore not only a question of model capability or training efficiency. It is a question of whether the resulting geometry supports the interpretability and control methods that will be applied to the model after training.

Limitations. Our models are small (45.4M parameters, 6 layers). Whether the conditioning patterns persist at the scale of deployed language models is an open question. The steering evaluation uses gendered word pairs as the primary concept; the downstream validation covers four tasks but does not include safety-specific evaluations such as refusal or toxicity. The 0.3 cosine threshold is empirical and may shift with model scale or domain.

Future work. Whether the conditioning improvement arises from limiting cumulative coordinate distortion across layers is a natural question. Appendix A states this as a conjecture. A direct test would require measuring the coordinate distortion (e.g., via the Jacobian of the layer map) and correlating it with Hessian conditioning. Extending the measurement to larger models and to safety-specific steering tasks would strengthen the practical relevance of the cosine diagnostic.

References

Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations (Workshop)*, 2017.

Shun-ichi Amari. *Information Geometry and Its Applications*. Springer, 2016.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.03819.

Jordan Frécon, Gilles Gasso, Massimiliano Pontil, and Saverio Salzo. Bregman neural networks. In *International Conference on Machine Learning*, pages 6779–6792. PMLR, 2022.

J. Clayton Kerce and Alexis Fox. The dual-stream transformer: Channelized architecture for interpretable language modeling. *arXiv preprint arXiv:2603.07461*, 2026.

Yizhou Liu, Ziming Liu, and Jeff Gore. Superposition yields robust neural scaling. In *Advances in Neural Information Processing Systems*, 2025. Best Paper Runner-up. arXiv:2505.10465.

Kiho Park, Todd Nief, Yo Joong Choe, and Victor Veitch. The information geometry of softmax: Probing and steering. *arXiv preprint arXiv:2602.15293*, 2026.

Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *2007 15th European Signal Processing Conference (EUSIPCO)*, pages 606–610, 2007.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.

Zhenhong Zhou, Haiyang Yu, Xinghua Zhang, Rongwu Xu, Fei Huang, Kun Wang, Yang Liu, Junfeng Fang, and Yongbin Li. On the role of attention heads in large language model safety. In *International Conference on Learning Representations*, 2025. Oral. arXiv:2410.13708.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A The rigidity conjecture

A standard transformer computes residual updates $x^{(\ell+1)} = x^{(\ell)} + f_\ell(x^{(\ell)})$. The map f_ℓ includes attention, a feed-forward network with nonlinear activations, skip connections, and layer normalization. Together these operations can rotate, rescale, shear, and nonlinearly warp the coordinate system from one layer to the next. At an intermediate layer ℓ, the effective Bregman metric depends on how the remaining layers $\ell+1, \ldots, L$ transform coordinates before reaching the output softmax. If the downstream layers distort the coordinate system substantially, the primal direction at layer ℓ (a straight line in layer-ℓ coordinates) will not correspond to a dual direction in the output geometry. The Hessian at layer ℓ will appear poorly conditioned because the coordinate system is misaligned with the output Bregman geometry.

The CASCADE architecture separates the representation into a frozen token stream x_t and a contextual stream x_e. The logits at any layer come from $x_t + x_e^{(\ell)}$. Because x_t is constant across layers, it anchors the combined representation. The contextual stream x_e is still subject to the full nonlinear layer map, but the fixed x_t component limits how far the total coordinate system can drift from the output geometry.

Conjecture 1 (Coordinate rigidity)**.** *Stream separation improves Bregman conditioning at intermediate layers primarily by limiting the cumulative coordinate distortion between layers. The frozen token stream anchors the coordinate system, reducing the misalignment between the layer-ℓ primal directions and the output-layer dual directions.*

Remark 1 (Unit consistency)**.** *An analogy to dimensional analysis in physics makes the conjecture more concrete. In physics, adding two quantities requires that they have the same units. The output softmax reads logits through the tied embedding matrix $\mathbf{W}_E^\top$ (the same matrix that maps tokens to embeddings, used in reverse to map representations back to vocabulary scores), so the natural "units" of the output are embedding-space coordinates. In CASCADE, the token stream $x_t = \mathbf{W}_E[t]$ is always in these units. The contextual stream $x_e^{(\ell)}$ must produce a vector addable to x_t in the same coordinate system, so unit consistency is maintained by construction at every layer. In a single-stream transformer, the representation $x^{(\ell)}$ passes through LayerNorm (which rescales), MLPs (which warp nonlinearly), and several composed layers. Nothing anchors it to the original embedding-space coordinates. Applying $\mathbf{W}_E^\top x^{(\ell)}$ at an intermediate layer computes a dot product between embedding-space vectors and a vector that may no longer be in embedding-space units. The per-layer auxiliary loss adds $\mathbf{W}_E^\top x^{(\ell)}$ at every layer, which acts as a soft enforcement of unit consistency through gradient signal. It helps, but the architecture still permits the representation to drift between loss computations. CASCADE enforces unit consistency structurally.*

The auxiliary per-layer loss places a softmax head at every layer and computes a cross-entropy loss against the target. This adds gradient signal encouraging each layer's representations to be useful for prediction, but it does not constrain the coordinate system. Each layer's auxiliary head can learn its own projection to compensate for coordinate drift.

Our results are consistent with the conjecture. A direct test would require measuring the coordinate distortion across layers (e.g., via the Jacobian of the layer map) and correlating it with Hessian conditioning.

B Architecture specification

Full details of the dual-stream transformer are given in Kerce and Fox [2026]. We summarize the elements relevant to this paper.

B.1 Dual-stream decomposition

The residual stream at layer ℓ is decomposed as

$$\mathbf{x}^{(\ell)} = \mathbf{x}_t^{(\ell)} + \mathbf{x}_e^{(\ell)},$$

where $\mathbf{x}_t^{(0)} = \mathbf{W}_E[t]$ is the token embedding and $\mathbf{x}_e^{(0)} = \mathbf{0}$.

B.2 CASCADE mode (frozen token stream)

In CASCADE mode the token stream is never updated:

$$\mathbf{x}_t^{(\ell)} = \mathbf{x}_t^{(0)} \quad \forall\, \ell, \tag{2}$$

$$\mathbf{x}_e^{(\ell+1)} = \mathbf{x}_e^{(\ell)} + \mathrm{Attn}^{(\ell)}(\mathbf{x}^{(\ell)}) + \mathrm{FFN}^{(\ell)}(\mathbf{x}^{(\ell)}). \tag{3}$$

All updates accumulate in the contextual stream. The combined representation used for logits at any layer is $\mathbf{x}^{(\ell)} = \mathbf{x}_t^{(0)} + \mathbf{x}_e^{(\ell)}$.

In the single-stream baseline, a single residual stream carries both token and contextual information, as in a standard transformer.

B.3 Per-layer supervision

At each layer ℓ, logits are computed through a shared language modeling head with tied embeddings:

$$\mathbf{z}^{(\ell)} = \mathrm{LayerNorm}\big(\mathbf{x}^{(\ell)}\big)\, \mathbf{W}_E^{\top}.$$

The training loss combines the final cross-entropy with weighted auxiliary losses:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(\mathbf{z}^{(L-1)}, y) + \lambda \sum_{\ell=0}^{L-2} w_\ell\, \mathcal{L}_{\mathrm{CE}}(\mathbf{z}^{(\ell)}, y),$$

with $\lambda = 0.1$ and linear decay weights $w_\ell = (\ell + 1)/L$.

B.4 Model variants

Model	Stream sep.	Aux. loss	Params
CASCADE + aux. loss	✓	✓	45.4M
CASCADE control	✓	—	45.4M
Single-stream + aux. loss	—	✓	45.4M
Single-stream control	—	—	45.4M

All models use 6 layers, 6 heads, 516-dimensional embeddings, gated attention, the GPT-2 tokenizer (50,257 tokens), and are trained on the same data mixture for the same number of steps.